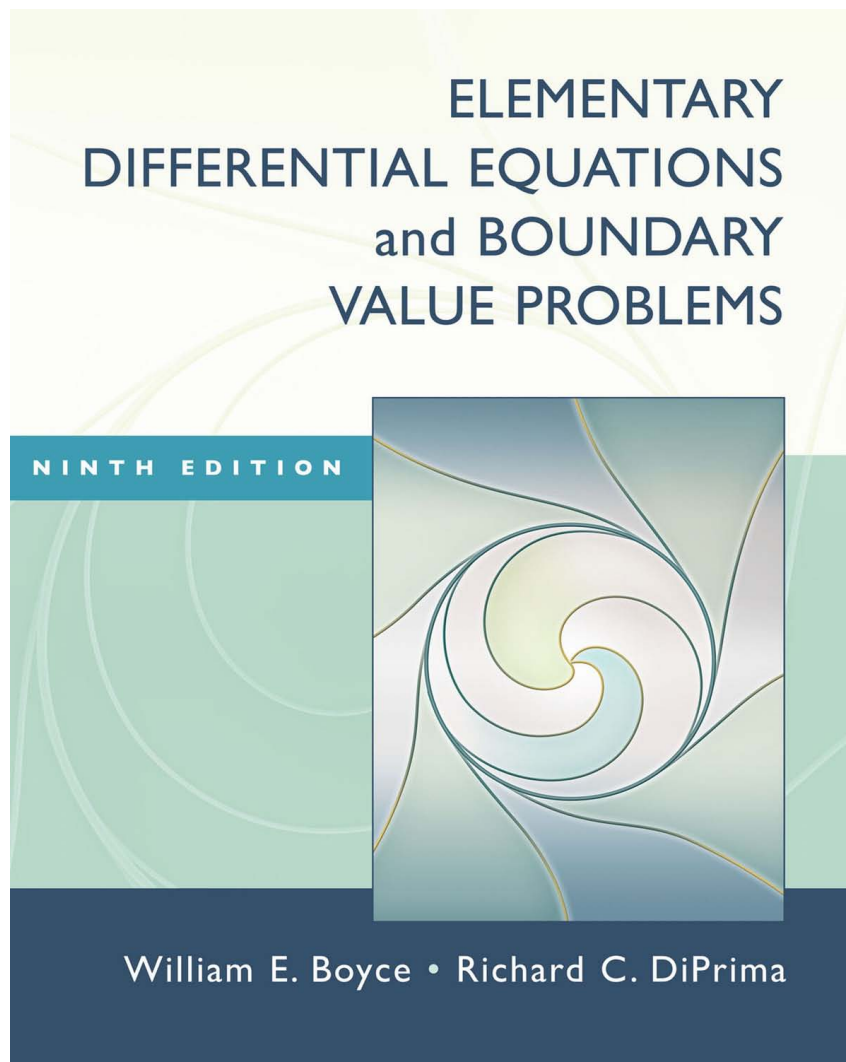


# Elementary Differential Equations and Boundary Value Problems - A Solution Manual

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# Chapter 1

## Introduction

### 1.1 Some Basic Mathematical Models; Direction Fields

## 1.2 Solutions of Some Differential Equations

1:

(a)

$$\begin{aligned}
 \frac{dy}{dt} &= -y + 5, \quad y(0) = y_0 \\
 -\frac{dy}{dt} &= -y + 5 \\
 -\frac{1}{y-5} \frac{dy}{dt} &= -1 \\
 \frac{d}{dt} \ln |y-5| &= -1 \\
 -\int \frac{d}{dt} \ln |y-5| dt &= -\int dt \\
 \ln |y-5| &= t + K_1 \\
 e^{\ln |y-5|} &= K_2 e^t \\
 \boxed{y(t) = 5 + K_2 e^t, \quad K_2 \neq 0}
 \end{aligned}$$

(b)

$$\begin{aligned}
 \frac{dy}{dt} &= -2y + 5, \quad y(0) = y_0 \\
 -\frac{1}{2} \frac{dy}{dt} &= y - \frac{5}{2} \\
 \frac{1}{y - \frac{5}{2}} \frac{dy}{dt} &= -2 \\
 \frac{d}{dt} \ln |y - \frac{5}{2}| &= -2 \\
 \int \frac{d}{dt} \ln |y - \frac{5}{2}| dt &= -2 \int dt \\
 \ln |y - \frac{5}{2}| &= -2t + K_1 \\
 e^{\ln |y - \frac{5}{2}|} &= K_2 e^{-2t} \\
 y(t) &= \frac{5}{2} + K_2 e^{-2t}
 \end{aligned}$$

Solving the initial value problem:

$$\begin{aligned}
 y_0 &= \frac{5}{2} + K_2 \quad \therefore \quad K_2 = y_0 - \frac{5}{2} \\
 \boxed{y(t) = \frac{5}{2} + \left(y_0 - \frac{5}{2}\right) e^{-2t}}
 \end{aligned}$$

(c)

$$\begin{aligned}
 \frac{dy}{dt} &= -2y + 10 \\
 \boxed{y(t) = 5 + (y_0 - 5)e^{-2t}}^1
 \end{aligned}$$

---

<sup>1</sup>The process itself is similar to the previous one in which the constant value is the only relevant modification. ■

**2:**

(a)

$$\begin{aligned}
\frac{dy}{dt} &= y - 5, \quad y(0) = y_0 \\
\frac{1}{y-5} \frac{dy}{dt} &= 1 \quad \therefore \quad \frac{d}{dt} \ln |y-5| = 1 \\
\int \frac{d}{dt} \ln |y-5| dt &= \int dt \quad \therefore \quad \ln |y-5| = t + K_1 \\
e^{\ln |y-5|} &= K_2 e^t \\
y(t) &= 5 + K_2 e^t, \quad K_2 \neq 0
\end{aligned}$$

Solving the initial value problem:

$$y_0 = 5 + K_2 \quad \therefore \quad K_2 = y_0 - 5$$

$$\boxed{y(t) = 5 + (y_0 - 5)e^t}$$

(b)

$$\begin{aligned}
\frac{dy}{dt} &= 2y - 5, \quad y(0) = y_0 \\
\frac{1}{2} \frac{dy}{dt} &= y - \frac{5}{2} \quad \therefore \quad \frac{1}{y - \frac{5}{2}} \frac{dy}{dt} = 2 \\
\frac{d}{dt} \ln \left| y - \frac{5}{2} \right| &= 2 \quad \therefore \quad \int \frac{d}{dt} \ln \left| y - \frac{5}{2} \right| dt = 2 \int dt \\
\ln \left| y - \frac{5}{2} \right| &= 2t + K_1 \\
e^{\ln \left| y - \frac{5}{2} \right|} &= K_2 e^{2t} \\
y(t) &= \frac{5}{2} + K_2 e^{2t}, \quad K_2 \neq 0
\end{aligned}$$

Solving the initial value problem:

$$y_0 = \frac{5}{2} + K_2 \quad \therefore \quad K_2 = y_0 - \frac{5}{2}$$

$$\boxed{y(t) = \frac{5}{2} + \left( y_0 - \frac{5}{2} \right) e^{2t}}$$

(c)

$$\begin{aligned}
\frac{dy}{dt} &= 2y - 10, \quad y(0) = y_0 \\
\frac{1}{2} \frac{dy}{dt} &= y - 5 \quad \therefore \quad \frac{1}{y-5} \frac{dy}{dt} = 2 \\
\frac{d}{dt} \ln |y-5| &= 2 \quad \therefore \quad \int \frac{d}{dt} \ln |y-5| dt = 2 \int dt \\
\ln |y-5| &= 2t + K_1 \\
e^{\ln |y-5|} &= K_2 e^{2t} \\
y(t) &= 5 + K_2 e^{2t}, \quad K_2 \neq 0
\end{aligned}$$

Solving the initial value problem:

$$y_0 = 5 + K_2 \quad \therefore \quad K_2 = y_0 - 5$$

$$\boxed{y(t) = 5 + (y_0 - 5)e^{2t}}$$

3:

(a)

$$\begin{aligned}
 \frac{dy}{dt} &= -ay + b \\
 \frac{1}{y - \frac{b}{a}} \frac{dy}{dt} &= -a \\
 \int \frac{d}{dt} \ln \left| y - \frac{a}{b} \right| dt &= -a \int dt \\
 \ln \left| y - \frac{a}{b} \right| &= -at + K_1 \quad \therefore \quad e^{\ln \left| y - \frac{a}{b} \right|} = K_2 e^{-at} \\
 \boxed{y(t) &= \frac{a}{b} + K_2 e^{-at}, \quad K_2 \neq 0}
 \end{aligned}$$

(b) This problem remains open.

4:

(a)

$$\frac{dy}{dt} = ay - b$$

A *equilibrium solution* consists of a solution that is constant in time. Be the differential equation

$$\frac{dy}{dt} = f(t, y).$$

A solution  $y(t) = c$ ,  $c \in \mathbb{R}$ , is called *equilibrium solutions* if satisfies

$$f(t, c) = 0, \quad \forall t \in D_f.$$

Therefore,

$$ay - b = 0 \quad \therefore \quad \boxed{y_e = \frac{b}{a}}.$$

(b)

$$\begin{aligned}
 Y(t) &= y - y_e \\
 &= \frac{b}{a} + K_2 e^{-at} - \frac{b}{a} \\
 Y(t) &= K_2 e^{-at}
 \end{aligned}$$

Therefore,

$$\boxed{Y(t) = K_2 e^{-at}, \quad K_2 \neq 0}.$$

5:

(a)

$$\begin{aligned}
 \frac{dy}{dt} &= ay - b \\
 \frac{dy}{dt} &= ay \quad \therefore \quad \int \frac{1}{y} dy = a \int dt \\
 \ln |y| &= at + K_1 \\
 e^{\ln |y|} &= K_2 e^{at} \\
 \boxed{y_1(t) &= K_2 e^{at}, \quad K_2 \neq 0}
 \end{aligned}$$

(b)

$$\begin{aligned}
\frac{dy}{dt} &= ay - b \quad \therefore \quad \frac{1}{y - \frac{b}{a}} \frac{dy}{dt} = a \\
\frac{d}{dt} \ln \left| y - \frac{b}{a} \right| &= a \\
\int \frac{d}{dt} \ln \left| y - \frac{b}{a} \right| dt &= a \int dt \\
\ln \left| y - \frac{b}{a} \right| &= 2t + K_1 \\
e^{\ln |y - \frac{b}{a}|} &= K_2 e^{at} \\
y(t) &= \frac{b}{a} + K_2 e^{at}, \quad K_2 \neq 0
\end{aligned}$$

Obtaining the constant  $K$ :

$$\begin{aligned}
y(t) &= y_1(t) + K \\
\frac{b}{a} + K_2 e^{at} &= K_2 e^{at} + K \\
\boxed{K} &= \frac{b}{a}
\end{aligned}$$

(c)

$$y = \frac{b}{a} + ce^{at}, \quad c \in \mathbb{R}$$

The method can be applied to any  $\frac{b}{a} = g(t)$ ,  $g : D_y \rightarrow \mathbb{R}^n$ . This depend to if is possible guess the shape that the solution will probably take.

■

6:

$$\begin{aligned}
\frac{dy}{dt} &= -ay + b \\
\boxed{y} &= -\frac{b}{a} + Ke^{-at}, \quad K \neq 0
\end{aligned}$$

■

7:

$$\begin{aligned}
\frac{dp}{dt} &= \frac{p-900}{2} \quad \therefore \quad \frac{1}{p-900} \frac{dp}{dt} = \frac{1}{2} \\
\int \frac{d}{dt} \ln |p-900| dt &= \frac{1}{2} \int dt \\
\ln |p-900| &= \frac{t}{2} + K_1 \\
e^{\ln |p-900|} &= K_2 e^{\frac{t}{2}} \\
p &= 900 + K_2 e^{\frac{t}{2}}, \quad K_2 \neq 0
\end{aligned}$$

(a) Solving que initial value problem, we obtain:

$$\begin{aligned}
p(0) &= 850 \\
850 &= 900 + K_2 \quad \therefore \quad K_2 = -50 \\
p(t) &= 900 - 50e^{\frac{t}{2}} \quad \therefore \quad e^{\frac{t}{2}} = 18 \\
\ln e^{\frac{t}{2}} &= \ln 18 \\
t = 2 \ln 18 \quad \therefore \quad \boxed{t \approx 5.78 \text{ months}}
\end{aligned}$$

(b)

$$\begin{aligned}
 p(0) &= p_0, \quad p_0 \in ]0, 900[ \\
 p_0 &= 900 + K_2 \quad \therefore \quad K_2 = p_0 - 900 \\
 p(t) &= 900 + (p_0 - 900)e^{\frac{t}{2}} \quad \therefore \quad \frac{900}{900 - p_0} = e^{\frac{t}{2}}
 \end{aligned}$$

$$\ln e^{\frac{t}{2}} = \ln 900 - \ln(900 - p_0)$$

$$t = 2 \ln \frac{900}{900 - p_0}, \quad p_0 \in ]0, 900[$$

(c)

$$\ln \frac{900}{900 - p_0} = \frac{t}{2}$$

$$e^{\ln \frac{900}{900 - p_0}} = e^{-\frac{t}{2}}$$

$$\frac{900}{900 - p_0} = e^{-\frac{t}{2}}$$

$$p_0 = 900(1 - e^{\frac{t}{2}})$$

$$p_0 = 1 - e^{-6} \quad \therefore \quad \boxed{p_0 \approx 0.9975}$$

Verify the problem to confirm the answer.

■

8:

$$\frac{dp}{dt} = rp$$

(a)

$$\int \frac{dp}{p} = r \int dt \quad \therefore \quad \ln |p| = rt + K_1$$

$$e^{\ln |p|} = K_2 e^{rt}$$

$$p(t) = K_2 e^{rt}, \quad K_2 \neq 0$$

Using the initial values:

$$p(0) = K_2$$

$$P(30) = p(0)e^{rt}$$

$$2p(0) = p(0)e^{rt}$$

$$e^{30r} = 2 \quad \therefore \quad \ln e^{30r} = \ln 2$$

$$30r = \ln 2$$

$$\boxed{r = \frac{\ln 2}{30}}$$

(b)

$$p(t) = p(0)e^{rt}$$

$$p(N) = p(0)e^{Nr}$$

$$2p(0) = p(0)e^{Nr} \quad \therefore \quad \ln e^{Nr} = \ln 2$$

$$Nr = \ln 2 \quad \therefore \quad \boxed{r = \frac{\ln 2}{N}}$$



9:

$$\frac{dv}{dt} = 9.8 - \frac{v}{5}, \quad v(0) = 0$$

- (a) The limit velocity corresponding to that at which there is no further variation in this module, satisfying

$$\begin{aligned} \frac{dv}{dt} &= 0 \\ 9.8 - \frac{v}{5} &= 0 \quad \therefore \quad v_{\infty} = 49 \text{ m/s.} \end{aligned}$$

The required velocity is, therefore,  $v(t_0) = 0.98v_{\infty}$ .

$$\begin{aligned} -\frac{dv}{dt} &= \frac{v-49}{5} \quad \therefore \quad \frac{1}{v-49} \frac{dv}{dt} = -\frac{1}{5} \\ \frac{d}{dt} \ln |v-49| &= -\frac{1}{5} \\ \int \frac{d}{dt} \ln |v-49| dt &= -\frac{1}{5} \int dt \\ \ln |v-49| &= -\frac{t}{5} + K_1 \quad \therefore \quad e^{\ln |v-49|} = K_2 e^{-\frac{t}{5}} \\ v(t) &= 49 + K_2 e^{-\frac{t}{5}} \end{aligned}$$

Be  $v(t_0) = 0.98v_{\infty}$ ,  $v_{\infty} = 49 \text{ m/s}$ :

$$\begin{aligned} e^{-\frac{t_0}{5}} &= 1 - \frac{0.98v_{\infty}}{49} \quad \therefore \quad \ln e^{-\frac{t_0}{5}} = \ln \left[ 1 - \frac{0.98v_{\infty}}{49} \right] \\ t_0 &= -5 \ln \left[ 1 - \frac{0.98v_{\infty}}{49} \right] \\ t_0 &\approx 19.56 \text{ s} \end{aligned}$$

(b)

$$\begin{aligned} v(t) &= 49(1 - e^{-\frac{t}{5}}) \\ \frac{dx}{dt} &= 49(1 - e^{-\frac{t}{5}}) \quad \therefore \quad \int dx = 49 \int (1 - e^{-\frac{t}{5}}) dt \\ x(t) &= 49t + 245e^{-\frac{t}{5}} + K_1 \\ x(t) &= 49t + 245e^{-\frac{t}{5}} - 245^2 \\ x(t_0) &= 49t_0 + 245e^{-\frac{t_0}{5}} - 245 \quad \therefore \quad x(t_0) \approx 718.34 \text{ m} \end{aligned}$$

10:

$$\frac{dv}{dt} = 9.8$$

- (a) Be  $v(0) = 0$  the initial value problem.

$$\begin{aligned} \int dv &= 9.8 \int dt \\ v(t) &= 9.8t + K_1 \end{aligned}$$

---

<sup>2</sup>We assume  $K_1 = -245$  since the fall starts from rest.

Solving the IVP:

$$v(0) = K_1 \quad \therefore \quad K_1 = 0$$

$$\boxed{v(t) = 9.8t}$$

(b) Be  $x(0) = 0$  the initial value problem.

$$\frac{dx}{dt} = 9.8t$$

$$\int dx = 9.8 \int t dt$$

$$x = 4.9t^2 + K_2$$

$$x(t) = 4.9t^2$$

Using  $x(t_0) = 300$  as information to the problem.

$$300 = 4.9 t_0^2$$

$$\boxed{t_0 \approx 7.82 \text{ s}}$$

(c)

$$v(t_0) = 9.8 t_0$$

$$\boxed{v(t_0) \approx 76.68 \text{ m/s}}$$

**11:**

This problem remains open.

**12:**

$$\begin{cases} \frac{dQ}{dt} = -rQ, & r > 0 \\ Q(0) = 100, & Q(1) = 82.04 \end{cases}$$

(a)

$$\int \frac{dQ}{Q} = -r \int dt \quad \therefore \quad \ln |Q| = -rt + K_1$$

$$e^{\ln |Q|} = K_2 e^{-rt}$$

$$Q(t) = K_2 e^{-rt}, K_2 \neq 0$$

Be  $Q(0) = 100$ ,  $K_2 = 100$ :

$$Q(t) = 100e^{-rt} \quad \therefore \quad Q(1) = 100e^{-r}$$

$$e^{-r} = \frac{82.04}{100}$$

$$\ln e^{-r} = \ln \frac{82.04}{100}$$

$$r = -\ln \frac{82.04}{100} \quad \therefore \quad \boxed{r \approx 0.198}$$

(b)

$$\boxed{Q(t) = 100e^{-rt}}$$

(c)

$$\begin{aligned}
Q(t_0) &= \frac{Q(0)}{2} \quad \therefore \quad 100e^{-rt_0} = \frac{Q(0)}{2} \\
e^{rt_0} &= \frac{1}{2} \\
\ln e^{-rt_0} &= \ln \frac{1}{2} \\
t_0 &= -\frac{\ln \frac{1}{2}}{r} \quad \therefore \quad \boxed{t_0 \approx 3.5 \text{ weeks}}
\end{aligned}$$

13:

$$\begin{aligned}
\frac{dQ}{dt} &= -rQ & Q(\tau) &= Q(0)e^{-r\tau} \\
\frac{Q(0)}{2} &= Q(0)e^{-r\tau}
\end{aligned}$$

Solving the initial value problem,  $Q(0) = K_2$ :

$$\begin{aligned}
e^{-r\tau} &= \frac{1}{2} \\
\ln(-r\tau) &= \ln \frac{1}{2} \\
r\tau &= -\ln \frac{1}{2} = \ln \left( \frac{1}{2} \right)^{-1} \quad \therefore \quad \boxed{r\tau = \ln 2}
\end{aligned}$$

14:

$$\begin{aligned}
\frac{dQ}{dt} &= -rQ, & \begin{cases} Q(\tau) = \frac{Q(0)}{2} \\ Q(t_0) = \frac{Q(\tau)}{2} \end{cases} \\
\int \frac{dQ}{Q} &= -r \int dt \quad \therefore \quad \ln |Q| = -rt + K_1 \\
e^{\ln |Q|} &= K_2 e^{-rt} \\
Q(t) &= K_2 e^{-rt}, \quad K_2 \neq 0 \\
Q(t) &= Q(0) e^{-rt}
\end{aligned}$$

$$\begin{aligned}
Q(t_0) &= Q(0) e^{-rt_0} \\
\frac{Q(\tau)}{2} &= Q(0) e^{-rt_0} \\
e^{-rt_0} &= \frac{Q(\tau)}{2Q(0)} \\
\ln e^{-rt_0} &= \ln \frac{Q(\tau)}{2Q(0)} \quad \therefore \quad \boxed{t_0 = -\frac{\ln \frac{Q(\tau)}{2Q(0)}}{r}}
\end{aligned}$$

15:

$$\begin{cases} \frac{du}{dt} = -K(u - T), & K \geq 0 \\ u(0) = u_0 \end{cases}$$

(a)

$$\begin{aligned}
\frac{du}{dt} &= -K(u - T) \quad \therefore \quad \frac{1}{u - T} \frac{du}{dt} = -K \\
\frac{d}{dt} \ln |u - T| &= -K \\
\int \frac{d}{dt} \ln |u - T| dt &= -K \int dt \\
\ln |u - T| &= -Kt + C \\
e^{\ln |u - T|} &= ce^{-Kt} \\
u(t) &= ce^{-Kt} + T
\end{aligned}$$

Solving the initial value problem, we have that  $c = u_0 - T$ . Therefore:

$$\boxed{u(\tau) = (u_0 - T)e^{-K\tau} + T}.$$

(b) We assume that  $u(\tau) = \frac{u(0)}{2} = \frac{u_0 - T}{2}$ . Thus,

$$\begin{aligned}
\frac{u_0 - T}{2} &= (u - T)e^{-K\tau} + T \\
e^{-K\tau} &= \frac{u_0 - 3T}{2(u - T)} \\
\ln e^{-K\tau} &= \ln \frac{u_0 - 3T}{2(u - T)} \quad \therefore \quad \boxed{K = -\frac{\ln \frac{u_0 - T}{2(u - T)}}{\tau}}.
\end{aligned}$$

**16:**

Be  $u(0) = 70$ ,  $T = 10$ ,  $u(t_0) = 32$ . As seen in the **problem 15**:

$$\frac{du}{dt} = -K(u - T), \quad u(t) = ce^{-Kt} + T.$$

Solving the initial value problem,  $c = u(0) - T$ .

$$\begin{aligned}
u(t) &= (u(0) - T)e^{-Kt} + T \\
u(t_0) &= (u(0) - T)e^{-Kt} + T \quad \therefore \quad e^{-Kt} = \frac{u(t_0) - T}{u(0) - T}
\end{aligned}$$

Solving to  $t_0$ , we obtain:

$$t_0 = -\frac{\ln \frac{u(t_0) - T}{u(0) - T}}{K} \quad \therefore \quad \boxed{t_0 \approx 6.65 h}.$$

**17:**

$$R \frac{dQ}{dt} + \frac{Q}{C} = V$$

(a)

$$\begin{aligned}
\frac{dQ}{dt} &= \frac{V}{R} - \frac{Q}{RC} \quad \therefore \quad \frac{dQ}{dt} = \frac{1}{R} \left( \frac{VC - Q}{C} \right) \\
\frac{1}{Q - VC} \frac{dQ}{dt} &= -\frac{1}{R} \\
\int \frac{d}{dt} \ln |Q - VC| dt &= -\frac{1}{R} \int dt
\end{aligned}$$

$$\ln |Q - VC| = -\frac{t}{R} + C$$

$$e^{\ln |Q-VC|} = ce^{-\frac{t}{R}}$$

$$Q(t) = ce^{-\frac{t}{R}} + VC, \quad c = -VC \quad \therefore \quad \boxed{Q(t) = VC(1 - e^{-\frac{t}{R}})}$$

(b)

$$\begin{cases} \frac{dQ}{dt} = -\frac{Q-VC}{R} \\ \frac{dQ}{dt} = 0 \end{cases} \quad \therefore \quad \boxed{Q_L = VC}$$

(c)

$$\begin{cases} V(t_1) = 0 \\ Q(t_1) = Q_L = VC \end{cases} \quad \therefore \quad Q(t_1) = VC(1 - e^{-\frac{t_1}{R}})$$

$$Q(t_1) = VC(1 - e^{-\frac{t_1}{R}})$$

$$VC = VC(1 - e^{-\frac{t_1}{R}})$$

$$e^{-\frac{t_1}{R}} = 1$$

$$\ln e^{-\frac{t_1}{R}} = 0 \quad \therefore \quad \boxed{t_1 = 0}$$

■

18:

(a) Be  $V = 10^6$  gal the water volume of the lake. The rate substance inlet is given by:

$$r_{\text{inlet}} = 10^{-2} \frac{\text{g}}{\text{gal}} \cdot 300 \frac{\text{gal}}{\text{h}} = 3 \frac{\text{g}}{\text{h}}$$

The concentration of chemical product in the lake at any time  $t$  is given by:

$$\frac{Q(t)}{V} = \frac{Q(t)}{10^6 \text{ gal}}.$$

The exit rate is determined by:

$$r_{\text{exit}} = \frac{Q(t)}{10^6 \text{ gal}} \cdot 300 \frac{\text{gal}}{\text{h}}$$

$$r_{\text{exit}} = \frac{300 \frac{\text{gal}}{\text{h}}}{10^6 \text{ gal}} \cdot Q(t)$$

The model relates to the rate of chemical product in the lake is, therefore:

$$\frac{dQ}{dt} = r_{\text{exit}} - r_{\text{inlet}} \quad \therefore \quad \boxed{\frac{dQ}{dt} = 3[1 - 10^{-4}Q(t)]}$$

(b)

$$\frac{dQ}{dt} = 3[1 - 10^{-4}Q(t)]$$

$$\frac{10^4}{Q(t) - 10^4} \frac{dQ}{dt} = -3$$

$$10^4 \frac{d}{dt} \ln |Q(t) - 10^4| = -3$$

$$\int \frac{d}{dt} \ln |Q(t) - 10^4| dt = -3 \cdot 10^{-4} \int dt$$

$$\ln |Q(t) - 10^4| = -3 \cdot 10^{-4}t + C$$

$$e^{\ln |Q(t)-10^4|} = ce^{-3 \cdot 10^{-4}t} \quad \therefore \quad Q(t) = ce^{-3 \cdot 10^{-4}t} + 10^4$$

Be  $Q(0) = 0$ ,  $c = -10^4$ . Thus,

$$\begin{aligned} Q(t) &= 10^4(1 - e^{-3 \cdot 10^{-4}t}) \\ Q(8640) &= 10^4(1 - e^{-2.5}) \quad \therefore \quad \boxed{Q(8640) \approx 9179.15 \text{ g}}. \end{aligned}$$

(c)

$$\begin{aligned} \frac{dQ}{dt} &= -3 \cdot 10^{-4} Q(t) \\ \frac{1}{Q(t)} \frac{dQ}{dt} &= -3 \cdot 10^{-4} \\ \ln |Q| &= -3 \cdot 10^{-4}t + C \\ e^{\ln |Q|} &= ce^{-3 \cdot 10^{-4}t} \quad \therefore \quad \boxed{\begin{cases} Q(t) = ce^{-3 \cdot 10^{-4}t} \\ Q(0) = 9179.15 \end{cases}} \end{aligned}$$

(d) Be  $Q(0) = 0$ ,  $c = Q(0)$ . We have

$$\boxed{Q(t) = Q(0) e^{-3 \cdot 10^{-4}t}}.$$

(e)

$$\begin{aligned} \begin{cases} \frac{dQ}{dt} = Q(0) e^{-3 \cdot 10^{-4}t} \\ Q(t_0) = 10 \end{cases} &\quad \therefore \quad e^{-3 \cdot 10^{-4}t_0} = \frac{10}{Q(0)} \\ \ln e^{-3 \cdot 10^{-4}t_0} &= \ln \frac{10}{Q(0)} \\ t_0 &= -3 \cdot 10^4 \ln \frac{10}{Q(0)} \quad \therefore \quad \boxed{t_0 \approx 947.5 \text{ days}} \end{aligned}$$

■

**19:**

Be  $V = 6 \cdot 10^4 \text{ gal}$ ,  $m_{\text{dye}} = 5 \text{ kg}$ ,  $m_{\text{water}} = 200 \frac{\text{gal}}{\text{min}}$ .

(a)

$$\frac{dq}{dt} = -\frac{200 \frac{\text{gal}}{\text{min}}}{6 \cdot 10^4 \text{ gal}} q(t) \quad \therefore \quad \boxed{\begin{cases} \frac{dq}{dt} = -\frac{1}{300} q(t) \\ q(0) = 5 \end{cases}}$$

(b)

$$\begin{aligned} \frac{1}{q} \frac{dq}{dt} &= -\frac{1}{300} \\ \int \frac{d}{dt} \ln |q| dt &= -\frac{1}{300} \int dt \\ \ln |q| &= -\frac{1}{300}t + C \\ e^{\ln |q|} &= ce^{-\frac{1}{300}t} \\ q(t) &= ce^{-\frac{1}{300}t}, \quad c \in \mathbb{R}_+^* \end{aligned}$$

Solving the initial value problem, in which  $c = q(0) = 5$ :

$$\boxed{q(t) = 5e^{-\frac{1}{300}t}}.$$

(c)

$$\begin{aligned}
 q(t_0) &= 2 \cdot 10^{-2} \frac{\text{g}}{\text{gal}} \cdot 6 \cdot 10^4 \text{ gal} \\
 &= 1200 \text{ g} \\
 q(t_0) &= 1.2 \text{ kg}
 \end{aligned}$$

Using the given value,

$$\begin{aligned}
 q(t_0) &= 5e^{-\frac{1}{300}t_0} \\
 \ln e^{-\frac{1}{300}t_0} &= \ln 0.24 \\
 t_0 &= -300 \ln 0.24 \quad \therefore \quad \boxed{t_0 \approx 428.13 \text{ min}}.
 \end{aligned}$$

As  $t_0 \approx 7.13 \text{ h}$ ,  $t_0 > 4 \text{ h}$ , the system don't will can filter the water at the limit of 4 h.

(d) Be  $x$  the flow rate promoted by the water filter.

$$\begin{aligned}
 \frac{dq}{dt} &= -\frac{x \frac{\text{gal}}{\text{min}}}{6 \cdot 10^4 \text{ gal}} q(t) \\
 q(t_0) &= 2 \cdot 10^{-2} \frac{\text{g}}{\text{gal}} \cdot 6 \cdot 10^4 \text{ gal} = 1200 \text{ g} \frac{\text{kg}}{10^3 \text{ g}} = 1.2 \text{ kg} \\
 t_0 &= 4 \text{ h} \frac{60 \text{ min}}{\text{h}} = 240 \text{ min}
 \end{aligned}$$

$$\begin{aligned}
 q(t_0) &= 5e^{-\frac{x}{6 \cdot 10^4}t_0} \\
 1.2 &= 5e^{-4 \cdot 10^{-3}x} \\
 \ln e^{-4 \cdot 10^{-3}x} &= \ln 0.24 \\
 x &= -\frac{\ln 0.24}{4 \cdot 10^{-3}} \quad \therefore \quad \boxed{x \approx 356.78 \frac{\text{gal}}{\text{min}}}
 \end{aligned}$$

■

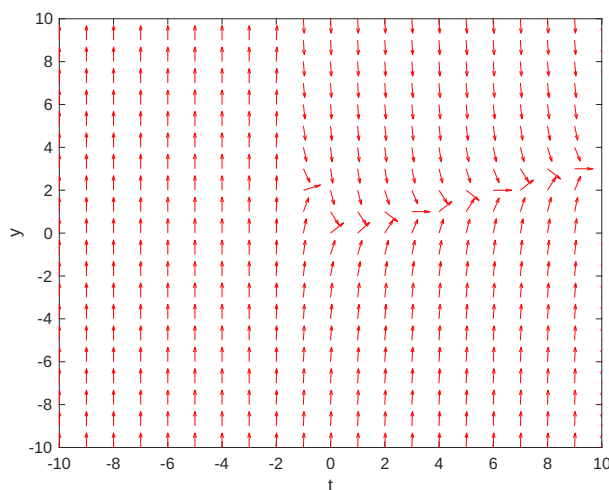
## Chapter 2

# First Order Differential Equations

### 2.1 Linear Equations; Method of Integrating Factors

1:

(a) Follows the direction field:



(b) The direction field is governed by the nullcline

$$y(t) = \frac{t + e^{-2t}}{3}.$$

As  $t \rightarrow +\infty$ , this curve has asymptotic slope of  $\frac{1}{3}$  and separates a lower branch where  $y' > 0$  from an higher branch where  $y' < 0$ . As  $t \rightarrow +\infty$  the exponential term dominates and the field points mostly upward, with the nullcline moving towards very large values of  $y$ .

(c)

$$\frac{dy}{dt} + 3y = t + e^{-2t}$$

Using the method os integrating factors, it follows that

$$\begin{aligned}\mu(t) &= e^{\int p(t) dt} = e^{\int 3 dt} \\ \mu(t) &= e^{3t}.\end{aligned}$$



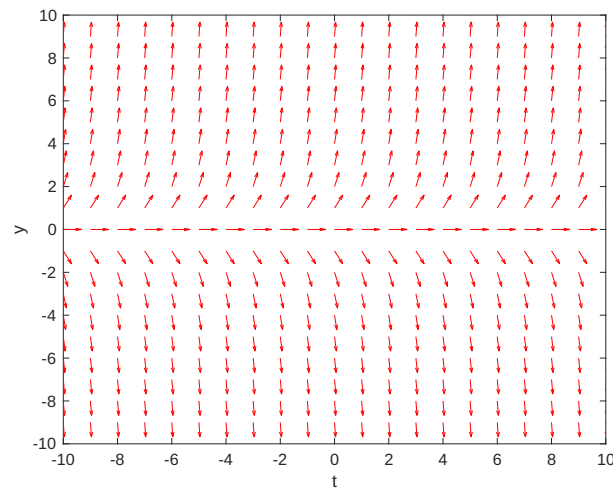
Thus,

$$\begin{aligned}\frac{d}{dt}(e^{3t}y) &= e^{3t}(t + e^{-2t}) \\ \int \frac{d}{dt}(e^{3t}y) dt &= \int te^{3t} + t dt \\ y(t) &= e^{-3t} \int te^{3t} + t dt + ce^{-3t} \\ y(t) &= e^{-3t} \left( \frac{te^{3t}}{3} - \frac{e^{3t}}{9} \right) + e^{-3t}(e^t) + ce^{-3t} \\ \boxed{y(t) &= \frac{t}{3} - \frac{1}{9} + e^{-2t} + ce^{-3t}}.\end{aligned}$$

■

2:

(a) Follows the direction field:



(b) For any  $y$  that remains limited (or grows more slowly than  $e^{2t}$ ), the forcing term  $t^2 e^{2t}$  dominates when  $t \rightarrow \infty$ . Thus,  $f(t, y) \rightarrow +\infty$  and the field segments become increasingly inclined upwards. Further, the equilibrium solution

$$\begin{aligned}2y + t^2 e^{2t} &= 0, \quad \frac{dy}{dt} = 0 \\ y(t) &= 0\end{aligned}$$

separates vector with opposite behaviors.

(c)

$$\frac{dy}{dt} - 2y = t^2 e^{2t}$$

Using the method of integrating factors, it follows that:

$$\begin{aligned}\mu(t) &= e^{\int p(t) dt} = e^{\int -2 dt} \\ \mu(t) &= e^{-2t}.\end{aligned}$$

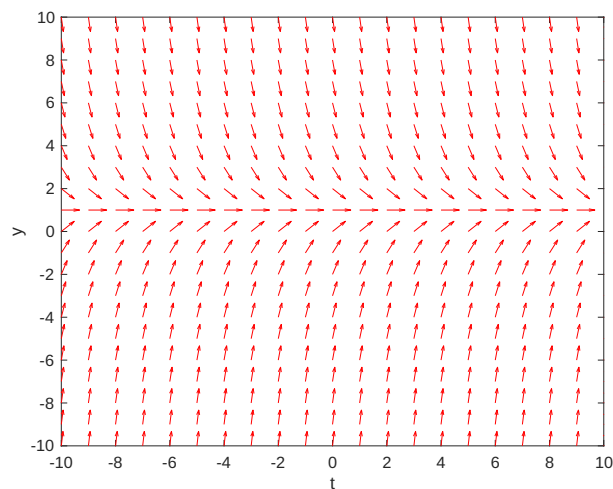
Thus,

$$\begin{aligned}\frac{d}{dt}(e^{-2t} y) &= t^2 \\ \int \frac{d}{dt}(e^{-2t} y) dt &= \int t^2 dt \\ y(t) &= e^{2t} \int t^2 dt + ce^{2t} \\ \boxed{y(t) &= \frac{t^3}{3} e^{2t} + ce^{2t}}.\end{aligned}$$

■

**3:**

(a) Follows the direction field:



(b) blablabla

(c)

$$\frac{dy}{dt} + y = t e^{-t} + 1$$

Using the method of integrating factors, it follows that:

$$\begin{aligned}\mu(t) &= e^{\int p(t) dt} = e^{\int dt} \\ \mu(t) &= e^t\end{aligned}$$

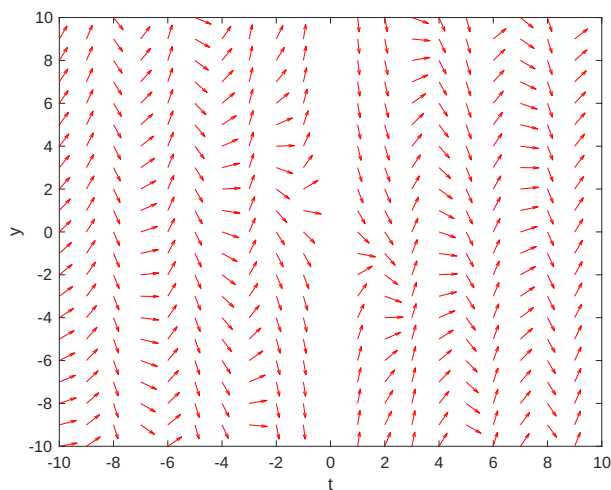
Thus,

$$\begin{aligned}\frac{d}{dt}(e^t y) &= t + e^t \\ \int \frac{d}{dt}(e^t y) dt &= \int t + e^t dt \\ y(t) &= e^{-t} \int t + e^t dt + ce^{-t} \\ \boxed{y(t) &= e^{-t} \frac{t^2}{2} + 1 + ce^{-t}}.\end{aligned}$$

■

**4:**

(a) Follows the direction field:



(b) blablabla

(c)

$$\frac{dy}{dt} + \frac{y}{t} = 3 \cos 2t$$

Using the method of integrating factors, it follows that:

$$\begin{aligned} \mu(t) &= e^{\int p(t) dt} = e^{\int \frac{dt}{t}} \\ &= e^{\ln |t|} \\ \mu(t) &= t. \end{aligned}$$

Thus,

$$\begin{aligned} \frac{d}{dt}(ty) &= 3t \cos 2t \\ \int \frac{d}{dt}(ty) dt &= \int 3t \cos 2t dt \\ y(t) &= t^{-1} \int 3t \cos 2t dt + ct^{-1} \end{aligned}$$

$$y(t) = \frac{3}{2} \sin 2t + \frac{3}{4} t^{-1} \cos 2t$$

■

5:

(a) Follows the direction field:

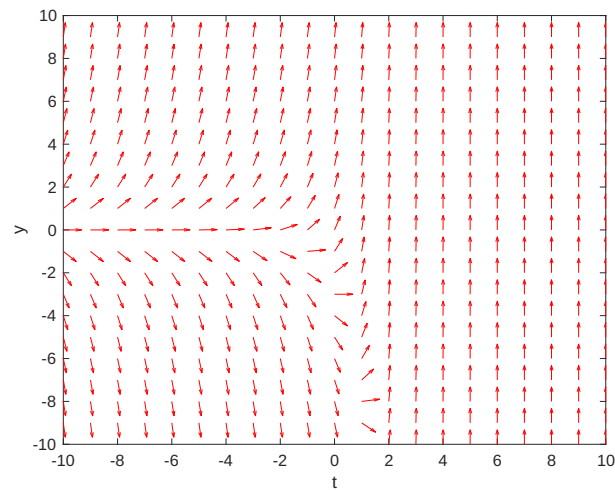
(b) blablabla

(c)

$$\frac{dy}{dt} - y = 3e^t$$

Using the method of integrating factors, it follows that:

$$\begin{aligned} \mu(t) &= e^{\int p(t) dt} = e^{-\int dt} \\ \mu(t) &= e^{-t} \end{aligned}$$



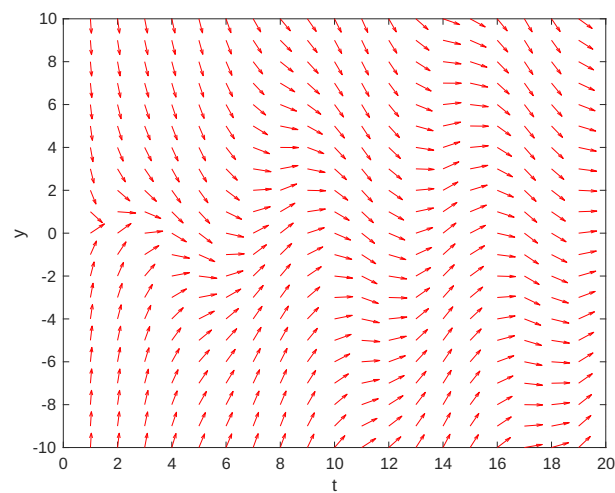
Thus,

$$\begin{aligned}\frac{d}{dt}(e^{-t}y) &= 3 \\ \int \frac{d}{dt}(e^{-t}y) dt &= \int 3 dt \\ y(t) &= e^t \int 3 dt + ce^t \\ \boxed{y(t) &= 3te^t + ce^t}\end{aligned}$$

■

6:

(a) Follows the direction field:



(b) blablabla

(c)

$$\begin{aligned}t \frac{dy}{dt} + 2y &= \sin t, \quad t > 0 \\ \frac{dy}{dt} + 2\frac{y}{t} &= \frac{\sin t}{t}\end{aligned}$$

Using the method of integrating factors, it follows that:

$$\begin{aligned}\mu(t) &= e^{\int p(t) dt} = e^{\int 2 \frac{dt}{t}} \\ \mu(t) &= t^2\end{aligned}$$

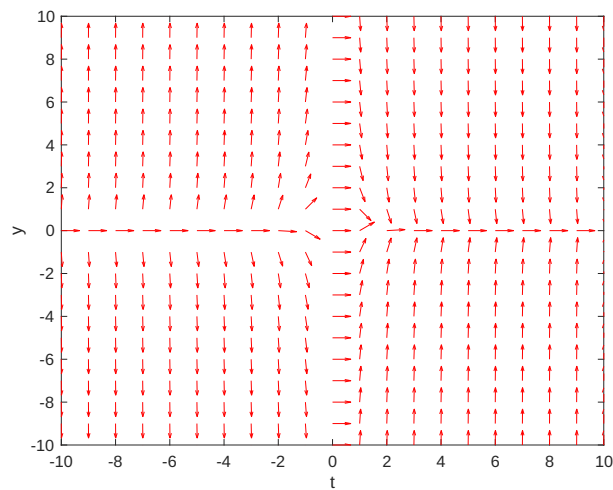
Thus,

$$\begin{aligned}\frac{d}{dt}(t^2 y) &= t \sin t \\ \int \frac{d}{dt}(t^2 y) dt &= \int t \sin t dt \\ y(t) &= t^{-1} \int t \sin t dt + ct^{-1} \\ \boxed{y(t) &= -\cos t + t^{-1} \sin t + ct^{-1}}\end{aligned}$$

■

7:

(a) Follows the direction field: Follows the direction field:



(b) f s d f d s f

(c)

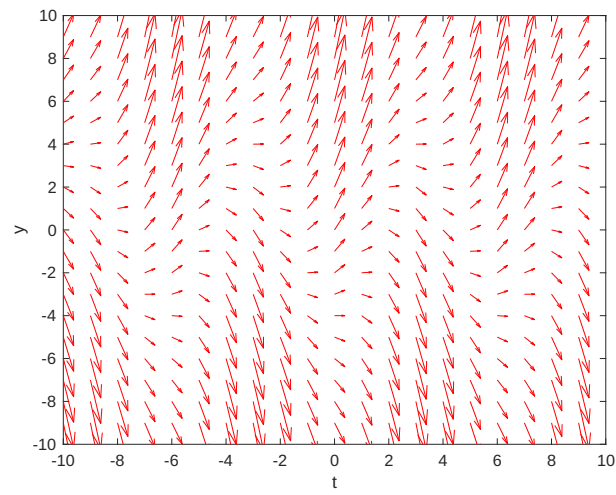
$$\frac{dy}{dt} + 2ty = 2te^{-t^2}$$

Using the method of integrating factors, it follows that:

$$\begin{aligned}\mu(t) &= e^{\int p(t) dt} = e^{\int 2t dt} \\ \mu(t) &= e^{t^2}\end{aligned}$$

Thus,

$$\begin{aligned}\frac{d}{dt}(e^{t^2} y) &= 2t \\ \int \frac{d}{dt}(e^{t^2} y) dt &= \int 2t dt \\ y(t) &= e^{-t^2} \int 2t dt + ce^{-t^2} \\ \boxed{y(t) &= t^2 e^{-t^2} + ce^{-t^2}}\end{aligned}$$



8:

(a) Teste para o problema 21

(b) asdfasd

## 2.2 Solutions of Some Differential Equations